HygionX - Bridging Artificial Intelligence and Explainable Medicine

- The aim of the project is to create a functional chatbot on healthcare, also binding the gaps, which most of the present chatbots were unable to do so

Existing Healthcare Chatbots & Apps:

- Babylon Health, Buoy Health, Ada Health {Symptom check & General healthcare}
- Woebot, Wysa, Replika {Mental Health & Emotional Support}
- Florence, Molly by Sensely {Medication & Chronic condition Support}
- Mayo clinic's; Cleaveland Clinic's AI Chatbot {Hospital & Provider specific chatbots}

Issues in existing ones:

- Doesnot integrate patient medical history before answering a question of patient
- Explainability of the given answer
- Trust issues of the answer given by the bot
- Single language centric like english
- Handling complex or rare cases, [diverting patient details to near hospitals with patient consent]
- Personalization of user data

Summary of challenges in Existing Chatbots:

- **Not truly personalized** →They don't combine full medical history, lifestyle, and wearable data into one view.
- **Reactive, not proactive** →They answer when asked but don't continuously monitor or predict problems early.
- **Limited trust & integration** →Explanations, empathy, and smooth connection with hospitals/EHRs are mostly missing.

For example, if I tell a bot like chatgpt, that I feel dizzy with headache, it give a list of possible reasons. It does not consider the previous user history, which can affect some individuals.

The course of action is that, while a user enters the chatbot, for the first time, the inbuilt mechanism prompts for some basic necessary details like previous surgeries, existing issues of user and all those.
So, whenever, the user gives a prompt or asks a question, then the bot retrieves all the important information about the user stored before by query retrieval mechanism, gives answer after consideration of all the issues, along wth the accuracy of its predicted answer. Due to this accuracy, the user will get to know about the genuinty of the model, and also, if the accuracy is high, only then the user may follow it. Also, the important details of the output text are highlighted by the presence of Explainable AI(XAI).

Requirements of the Project

1. Datasets:
    - PubMed
    - MIMIC III/IV [Requires Permit grants for usage, alternate ones can be searched]
    - Kaggle Healthcare QA Datasets
2. Databases:
    - SQL : Structured/Meta data [storing patient details, his past history in form of tables]
    - Elastic Search: Document Retrieval [Keyword search from user input in the model]
3. NLP/ML Models:
    - Frameworks:
        - HuggingFace
        - Pytorch [Backend]
    - Models:
        - BioBERT
        - ClinicalBERT
        - PubMedBERT [QA Task]
        - RAG Model Train [If all 3 of above fail to produce accurate results]
    - Fine Tuning Tools:
        - HuggingFace Trainer API
4. Explainable AI [XAI]:
    - Libraries:
        - SHAP [Token/Feature Level Contribution]
        - LIME [Local Model Explanation]
        - Captum [Interpretability]
    - Visualization:
        -Matplotlib
        - Seaborn
        - Plotly
5. Backend Framework:
    - FastAPI
    - Docker
6. Chatbot Layer:
    - Streamlit / Flask
7. Deployment:
    - Streamlit cloud / AWS [Free tier]

Improvements:

- BioBERT / PubMedBERT for Embeddings
- Elastic Search / FAISS for retrieval
- Light weight instruction tuned LLM [like LLAMA-2-Chat-Med, GPT-4….] for final answer synthesis
- If accuracy < threshold, escalate to "consult doctor" recommendation
- Anonymization + Encyption required for user data privacy and security
- Instead of raw SHAP/LIME plots, give reasons, considering the past data of user
- Updation of user data after each session
- Login and Authentication of user.
- Final phase of testing and validation: Check it through real doctors and medical students, so as to find any gaps left out, and for hallucinations

➢ Evaluation Benchmarks:
    - MedQA
    - MedMCQA
    - USMLE

➢ Hybrid DB:
    - SQL + NoSQL

Tentative Flow of Execution of Project:

- This Flowchart deals with the how the product works after completion, the step by step process.

User Login
↓
Check Consent & Retrieve User History (SQL DB)
↓
User Input (Symptom Query)
↓
Search Medical Knowledge Base (ElasticSearch / FAISS)
↓
Pass Data →ML Model (BioBERT/ClinicalBERT/PubMedBERT)
↓
Model Predicts Answer
↓
Explainability Layer Adds Reasons
↓
Confidence Check
-> If Confidence ≥ Threshold →Show Final Answer
-> If Confidence < Threshold →Suggest "Consult Doctor"
↓
Display Answer to User + Explanation
↓
Update User History (New symptoms/records)
↓
End

Execution Steps: [Tentative]

- This flowchart deals with the steps needed in the completion of the project.
- Some of the steps are optional and can be changed later according to the work flow,

Start
↓
Collect Datasets (PubMed, MIMIC, Kaggle)
↓
Preprocess Data (clean, tokenize, embeddings)
↓
Choose Models (BioBERT, ClinicalBERT, PubMedBERT, RAG)
↓
Fine-tune & Train Models (HuggingFace, PyTorch)
↓
Build Databases →SQL (patient history)
→ElasticSearch/FAISS (medical docs)
↓
Integrate Backend (FastAPI + Docker)
↓
Add Explainability Layer (SHAP, LIME, Captum)
↓
Design Chatbot UI (Streamlit / Flask)
↓
Add Security (Encryption, Login, Consent)
↓
Test & Validate with Doctors / Medical Students
↓
Deploy on Cloud (Streamlit Cloud / AWS Free Tier)
↓
End